



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories, demographics, and subscription behavior.



Dataset at a Glance

3,900 Rows

18 columns of transactional data

Demographics

Age, Gender, Location, Subscription

Purchases

Item, Category, Amount, Season

Behavior

Discounts, Frequency, Reviews

Data Preparation in Python

The image is a composite showing the data preparation workflow. On the left, a Python script for data cleaning is shown. In the center, a small table snippet is displayed. On the right, a screenshot of a PostgreSQL database window is shown.

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Database window screenshot showing a table with columns: Nom, A, Pnothelo, Bcoortip, D, Deciert. The table contains 20 rows of data.

→ Clean & Standardize

Imputed 37 missing Review Ratings; renamed columns to snake_case

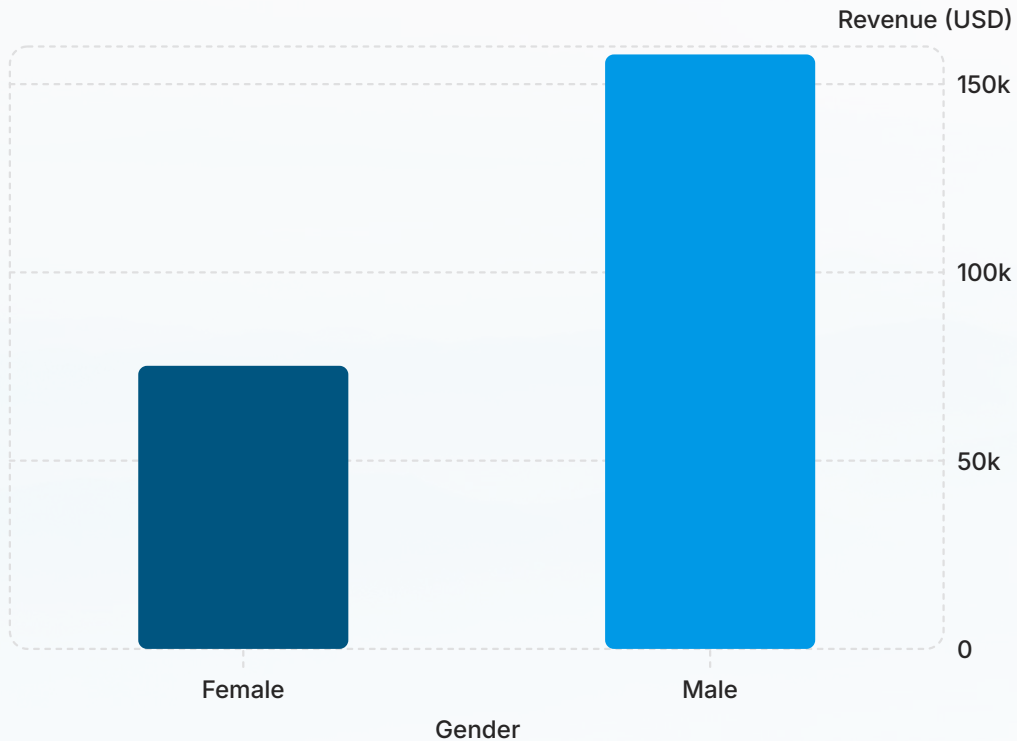
→ Feature Engineering

Created age_group and purchase_frequency_days

→ Database Load

Cleaned DataFrame loaded into PostgreSQL for SQL analysis

Revenue by Gender

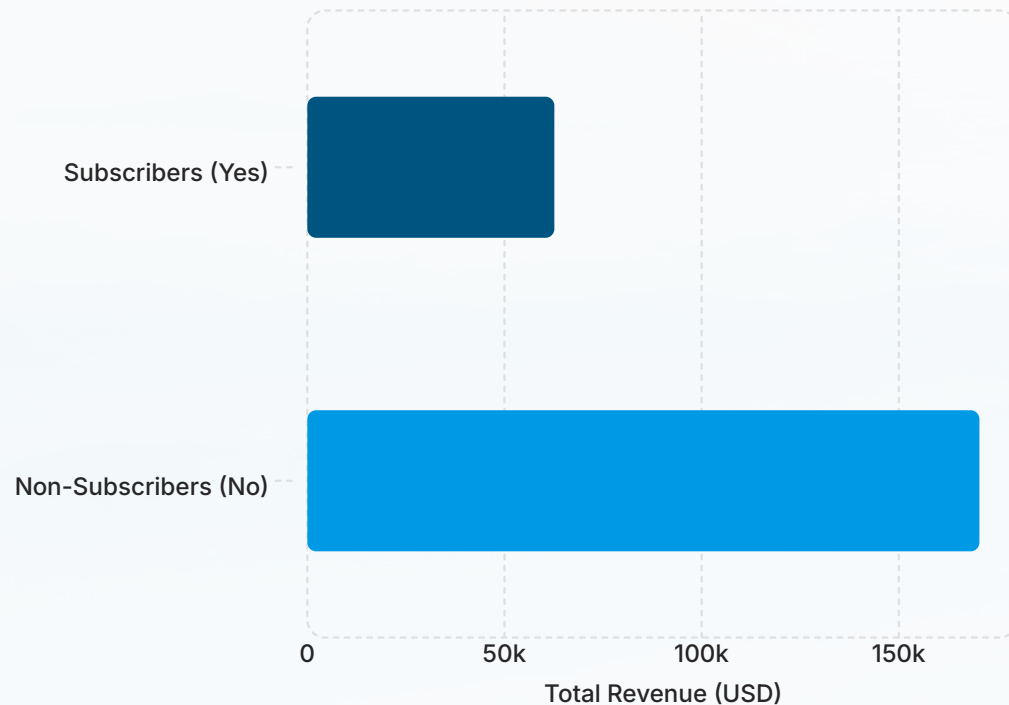


Key Insight

Male customers generated **2.1× more revenue** than female customers across 3,900 transactions.

Subscribers vs. Non-Subscribers

Subscription Status



27% Subscribers

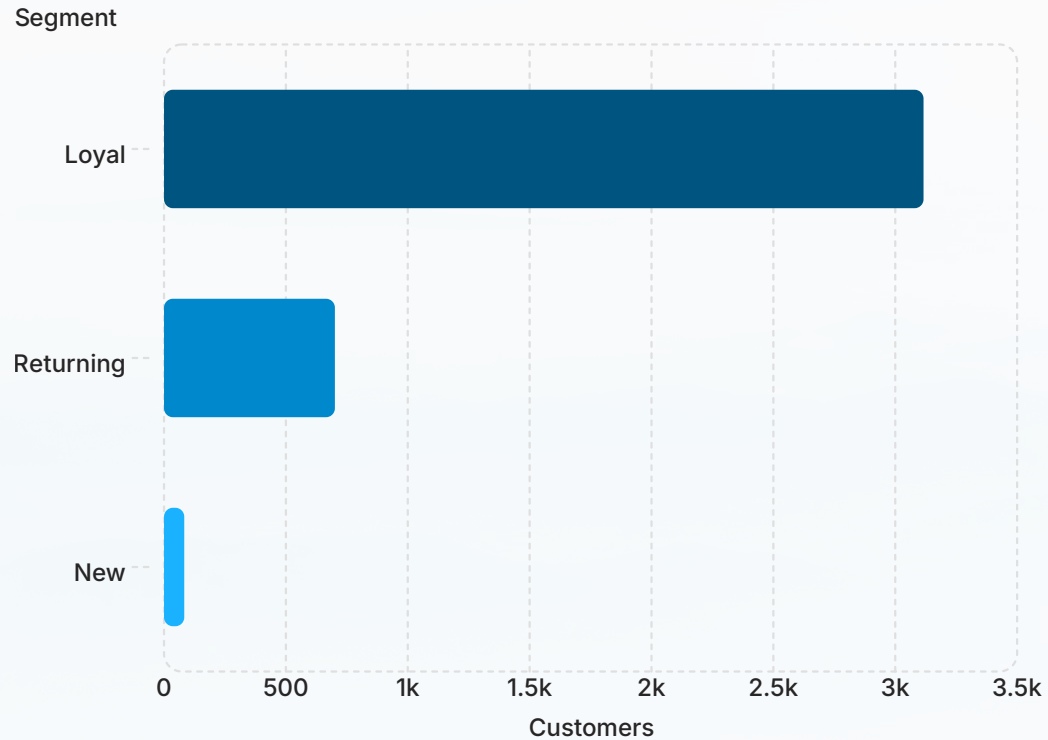
1,053 customers · Avg
spend \$59.49

73% Non-Subscribers

2,847 customers · Avg
spend \$59.87

Average spend is nearly identical — **subscription conversion is the growth lever.**

Customer Segmentation



Segment Breakdown

Loyal — 3,116

Core revenue drivers

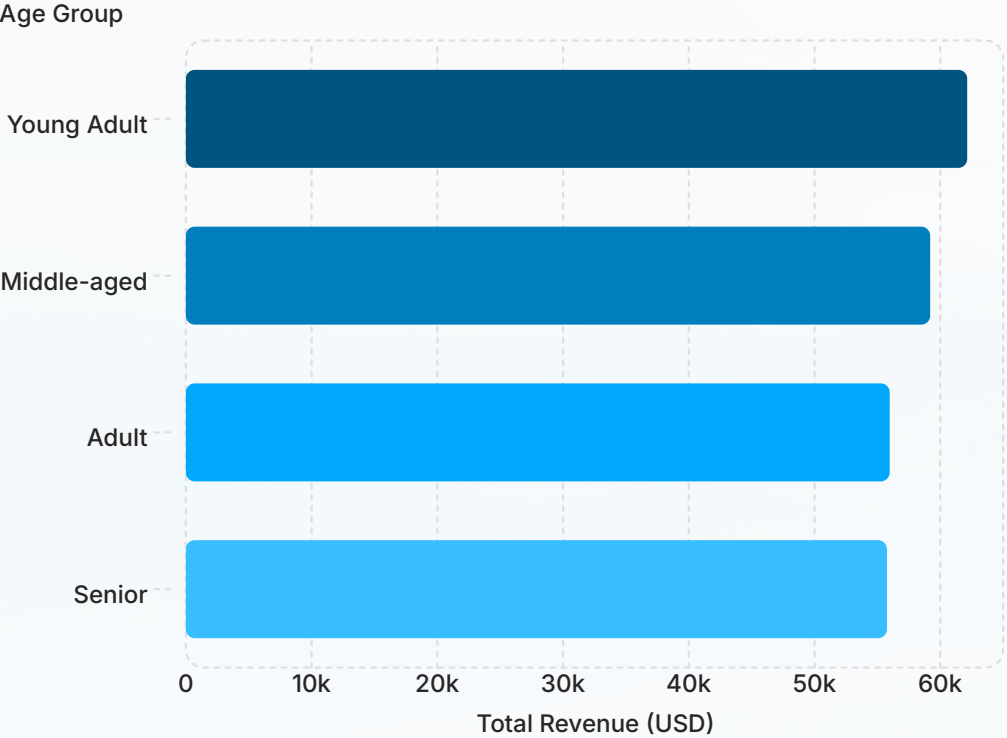
Returning — 701

Growth opportunity

New — 83

Acquisition focus

Revenue by Age Group



Key Insight

Young Adults lead in revenue contribution. Revenue is relatively evenly distributed across age groups — broad targeting is viable.

Top Products by Category



Clothing

Blouse · Pants · Shirt



Accessories

Jewelry · Sunglasses · Belt



Footwear

Sandals · Shoes · Sneakers



Outerwear

Jacket · Coat

Power BI Dashboard



Dashboard Highlights

01

3.9K Customers · Avg \$59.76 ·
3.75★

02

Filters: Subscription, Gender,
Category, Shipping

03

Revenue & Sales by Category
and Age Group

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discounts

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express shipping users

